Social Network Analysis

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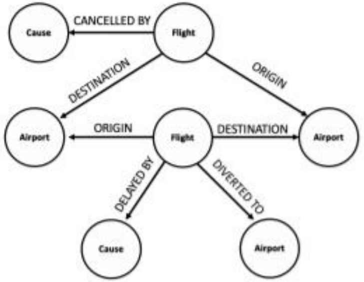
คณะสถิติประยุกต์ นิด้า

Social Network Analysis

- The process of investigating social structures through the use of networks and graph theory
  - ***nodes*** (individual actors, people, or things within the network) and
  - ***ties*, *edges*, or *links*** (relationships or interactions) that connect them.

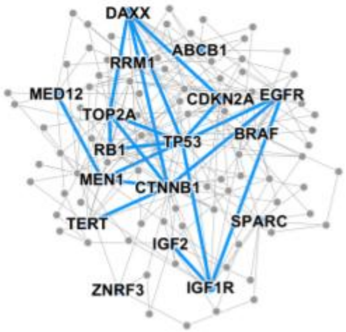
https://en.wikipedia.org/wiki/Social_network_analysis



Event Graphs

Image credit: SalientNetworks

Computer Networks



Disease Pathways

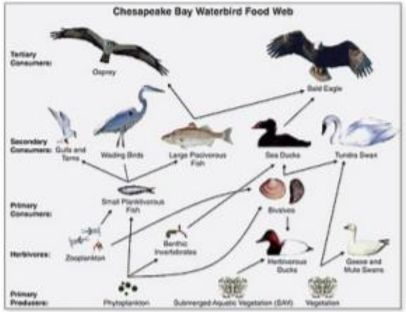


Image credit: Wikipedia

Food Webs

Image credit: Pinterest

Particle Networks

Image credit: visitlondon.com

Underground Networks

Image credit: Medium

Social Networks

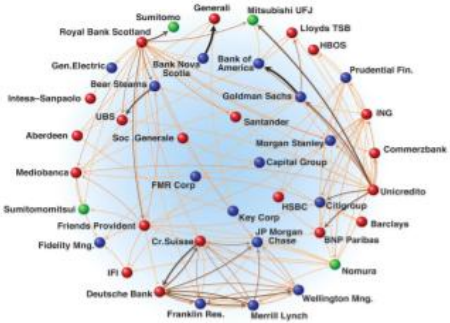


Image credit: Science

Economic Networks

Image credit: Lumen Learning

Communication Networks

Citation Networks

Image credit: Missoula Current News

Internet

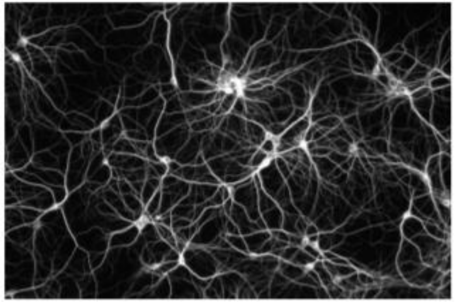
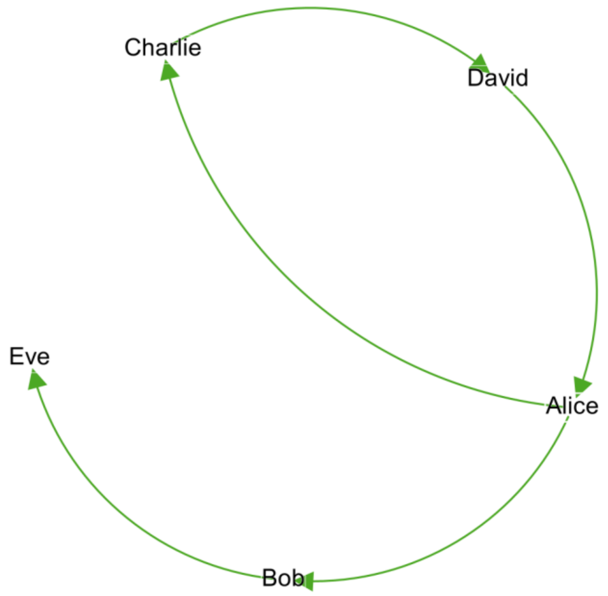


Image credit: The Conversation

Networks of Neurons

Why Network?

ID	Name	Department	Role
1	Alice	Marketing	Lead
2	Bob	Sales	Associate
3	Charlie	IT	Developer
4	David	Marketing	Creative
5	Eve	Sales	Coordinator



The Core Problem with Tabular Data -- The "Independence" Assumption

- Traditional tabular data is built on a massive assumption that every row is completely independent of the others.
- Tabular data looks at **attributes**. It tells you that Person A is 25, lives in Bangkok, and bought a specific product. It tells you Person B is 26, lives in Bangkok, and bought the same product.
- What it *cannot* tell you is that Person A bought the product **because** Person B recommended it to them. Tabular data misses the relationships, influence, and flow of information.

Comparison between tabular and network dataset

Feature	Tabular Dataset (Spreadsheets)	Network Dataset (Nodes & Edges)
Primary Focus	**Attributes:** What are the characteristics of this individual?	**Relationships:** How does this individual interact with others?
Structure	Rows (Entities) and Columns (Variables)	Nodes (Entities) and Edges (Connections)
Core Assumption	Entities act independently of one another.	Entities are interdependent and influence each other.
Best For...	Averages, demographics, revenue totals, individual scoring.	Viral spread, community building, finding hidden influencers, mapping echo chambers.
The Questions it Answers	*"Who spent the most money on our campaign?"*	*"Who convinced the most people to join our campaign?"*

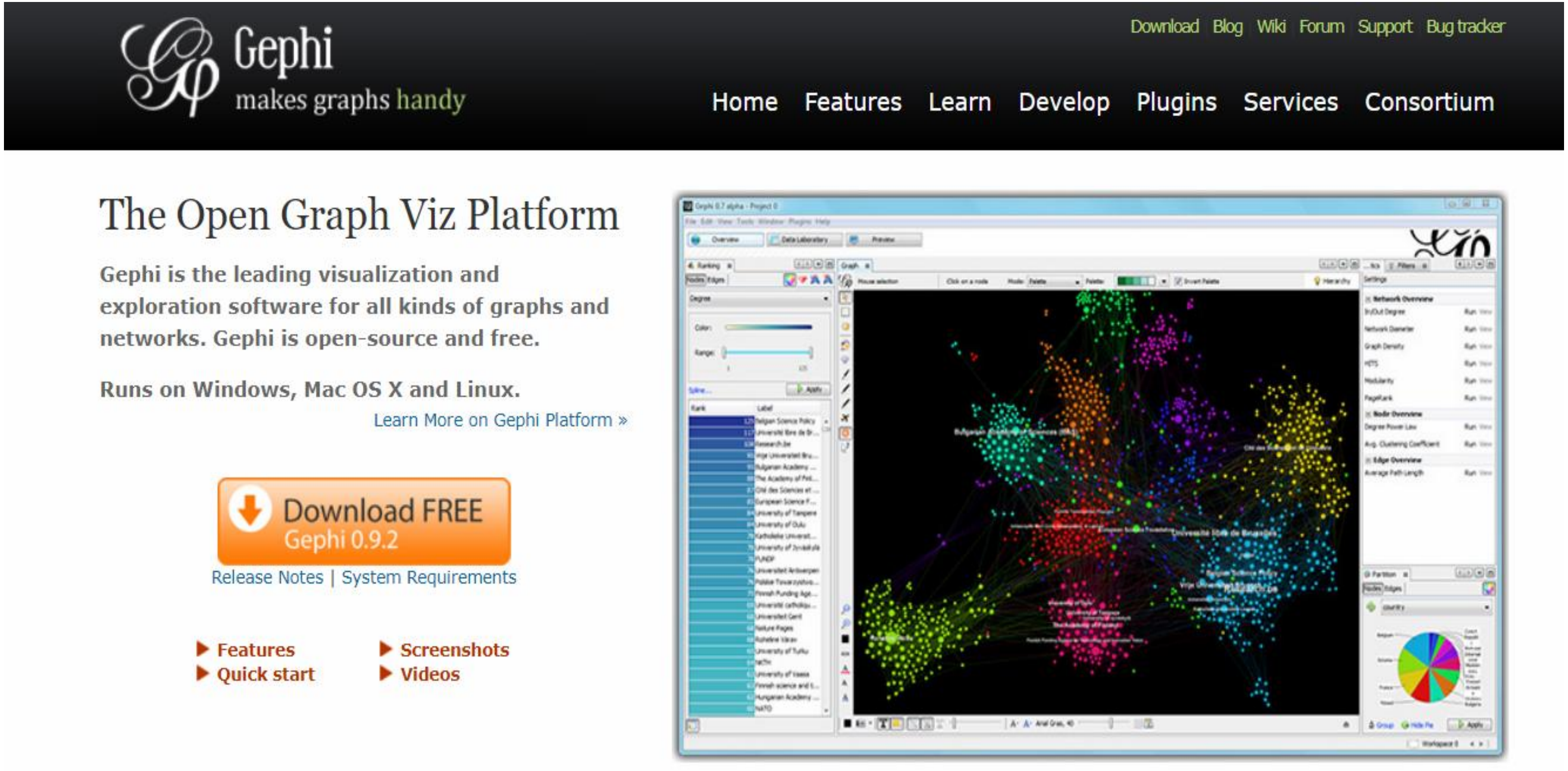
Tabular data focuses on isolation, while network data focuses on connection.

Network metrics

Metric	Description	Interpretation	Formula
Degree Centrality	Number of connections	Highly connected, popular, or influential	(number of connections) / (total nodes - 1)
Betweenness Centrality	Nodes that bridge other nodes	Bridge/Broker — controls information flow	(sum of all shortest paths through node) / (total shortest paths)
Closeness Centrality	Nodes closest to all others	Efficient spreader — reaches everyone quickly	(n-1) / sum of all shortest paths from node to all others
Eigenvector Centrality	A measure of a node's influence based on the quality of its connections — not just how many, but who they know.	Elite connector — connected to other important nodes	Eigenvector(node) = (1/λ) × Σ(neighbor's eigenvector scores) * Iterative calculation until convergence
PageRank	Influence scoring	Finding influential nodes, originally developed by Google for ranking web pages.	PR(A) = (1-d)/N + d × Σ(PR(Ti) / C(Ti)), Where d = damping factor (usually 0.85), N = total nodes, Ti = nodes linking to A, C(Ti) = out-links from Ti

Network metrics

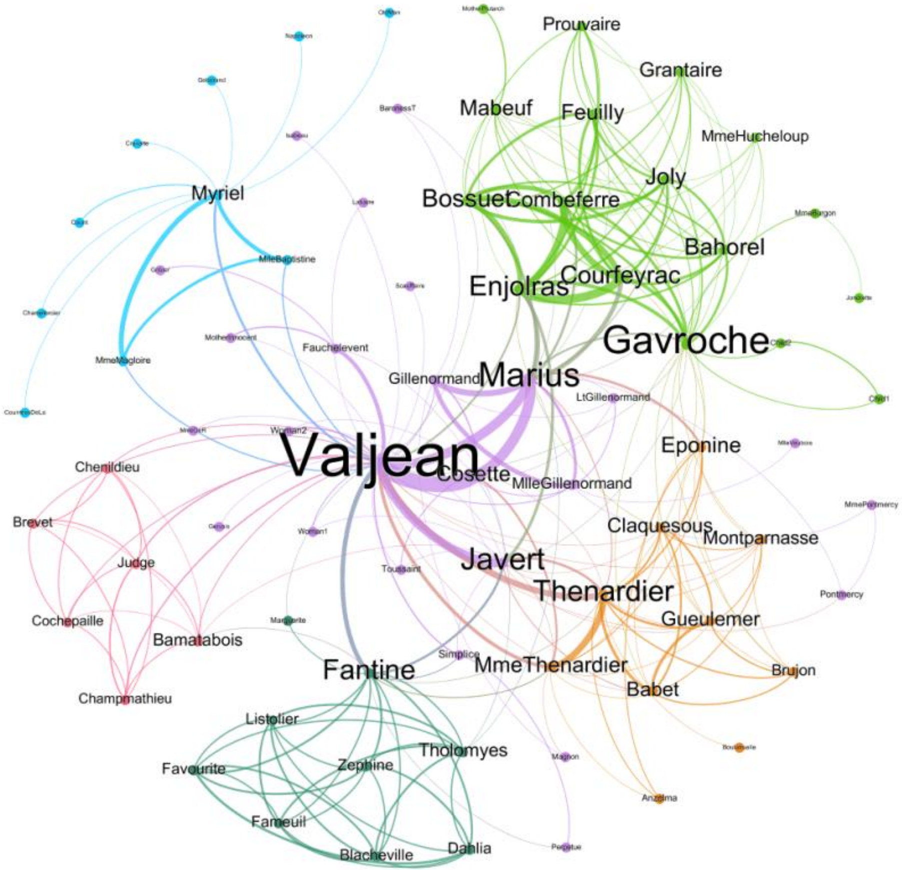
Metric	Description	Interpretation	Formula
Community Detection	The process of identifying groups of nodes that are more densely connected to each other than to the rest of the network.	A cluster of nodes with strong internal connections	Louvain, etc.
Path Analysis	The study of how nodes are connected through sequences of edges, focusing on the shortest routes between them.	Path, Shortest path, Average path	Dijkstra Source-Target Shortest Path.



- [Download]
- [Tutorial]

Use cases

	Jean Valjean
	Javert
	Fantine
	Cosette
	Thenardier
	Madame Thenardier
	Marius
	Enjorlas
	Eponine

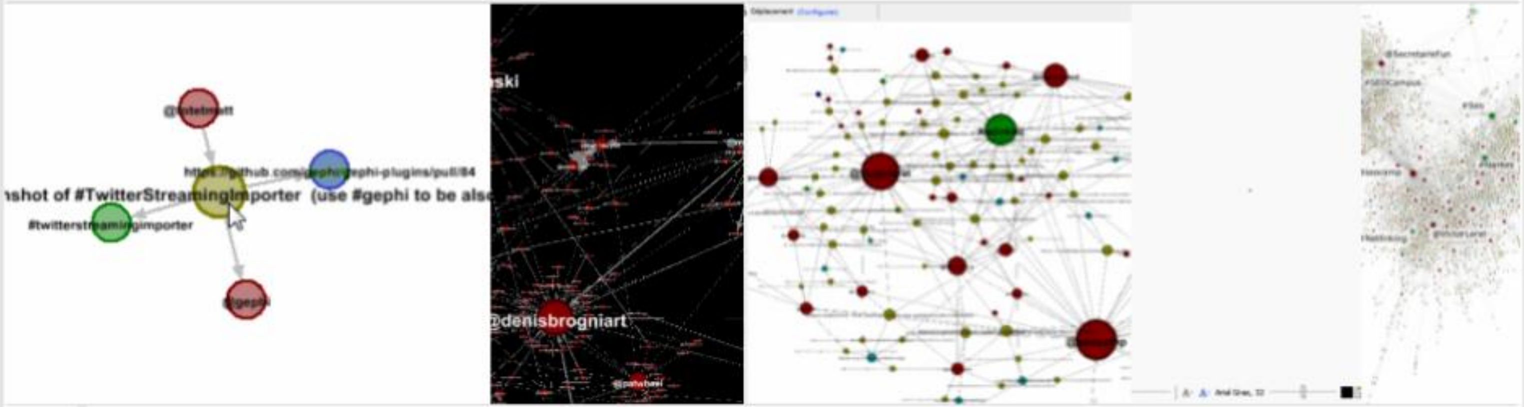


https://www.overthinkingit.com/2013/01/02/can-you-hear-the-people-graph-les-miserables/

TwitterStreamingImporter

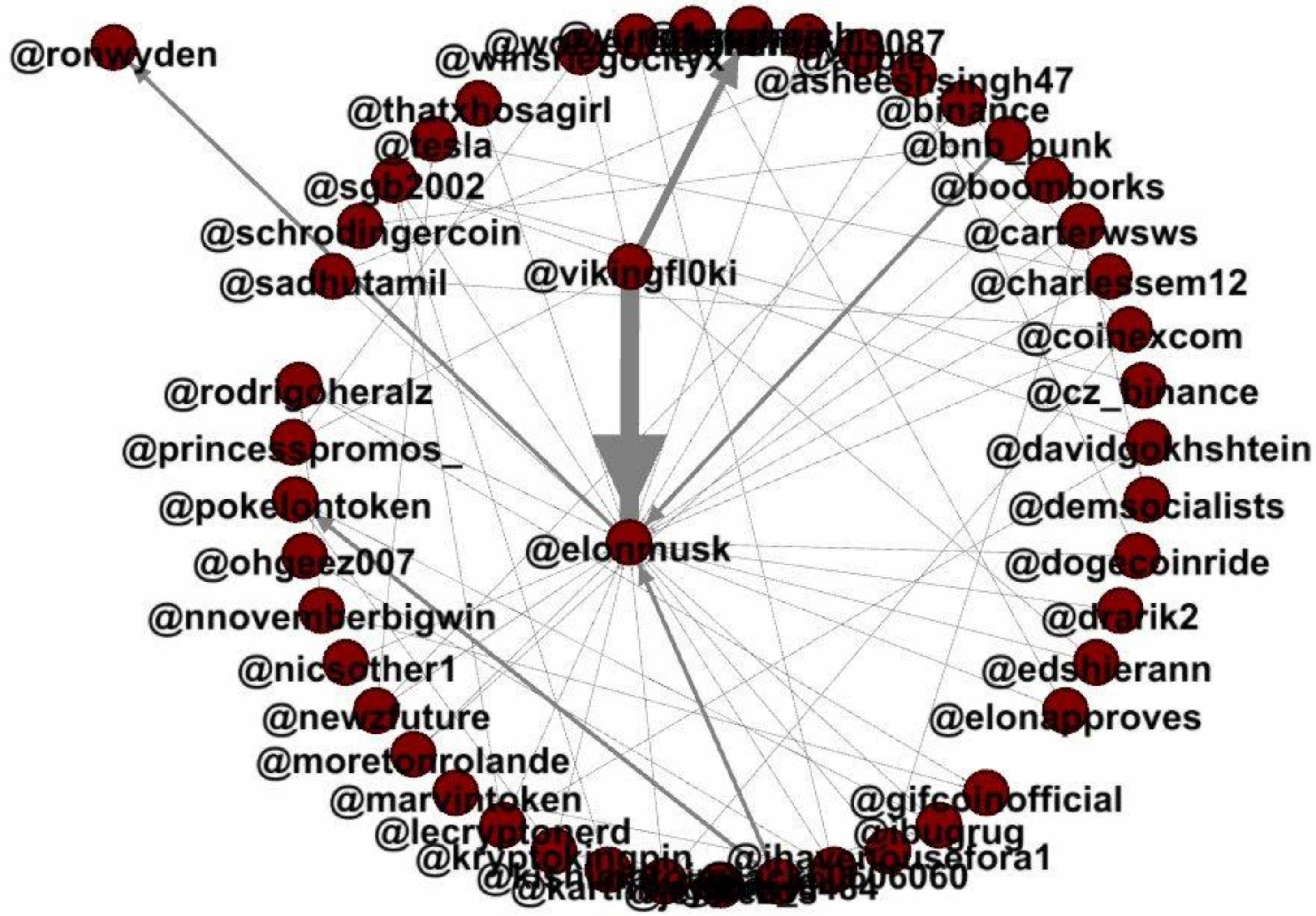
import

by @totetmatt



Realtime Twitter Graph Importer

https://gephi.org/plugins/#/plugin/twitter-streaming-importer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@pokelontoken
@ohgeez007
@nnovemberbigwin
@nicsother1
@newzfuture
@moretonrolande
@marvintoken
@lecryptonerd
@kryptokingpin
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Ron Wyden @RonWyden · Nov 7

Whether or not the world's wealthiest man pays any taxes at all shouldn't depend on the results of a Twitter poll. It's time for the Billionaires Income Tax.

> **Elon Musk** @elonmusk · Nov 7
>
> Much is made lately of unrealized gains being a means of tax avoidance, so I propose selling 10% of my Tesla stock.
>
> Do you support this?
>
> Show this poll

3.2K 2.2K 16.1K

Dataset

1. Simple network download
2. Food social download
3. Les Miserables download